IT4010E - INFORMATION SECURITY – SEM2023.1
FINAL EXAM: QUESTION-PROBLEM SHEET

PART A. *Answer 2 out of 4 following questions depending on your X value: For X=1,2,3 or 4 do 1-3, 1-4, 2-3 or 2-4, respectively*.

1) Discuss the concept of MAC code and then compare it with CRC-32 code.
2) Compare *chosen-plaintext attack* with *known-plaintext attack*.
3) Define and compare the concepts of *groups* and *roles* in access control.
4) Show the meaning and purpose of CAPTCHA and how it works.

PART B. *Solve just problem 1 or 2 bellow if your Y value is 1 or 2, respectively*.

1. For calculating a modulo inverse, a student has penned a GCD dry-run table as below but the second row has been accidentally ruined because of some water drop. Recover the full table and state the two numbers that are given to the modulo inverse calculation.

n_1	n_2	r	q	a_1	b_1	a_2	b_2
?	?	?	?	?	?	?	?
11	4	3	2	0	1	1	–3
4	3	1	1	1	–3	–2	7
2	1	0	3	–2	7	3	–10

2. For generating large prime numbers for her first try to create an RSA public key system, student Alice had sketched an algorithm with the main steps as followed:

   Step 1. Select a random odd number ***K*** from the range 1..*N* where $N=10^{200}$.
   Step 2. While *K* not is a prime Do $K \leftarrow K+2$
   Step 3. Let ***p***=*K* and Repeaf 1)-2) again but for generating the second prime ***q***.
   Done.

   a) Class TA Cathy however told Alice that this procedure was not quite good enough (resulting in an RSA system not secure enough) and asked Alice to fix a small detail to make it better. Can you find what was the problem that Cathy asked to fix?
   b) Alice intended to use an open source code for primality testing, namely algorithm PrimTest which could take roughly 0.15ms to decide if a 200-digit number is a prime or not. Using PrimTest code, can you roughly estimate how much time it would take in average to produce ***p*** & ***q*** by the above 3-step algorithm?

PART C. *All students are expected to work through the whole part*.

In a security class, instructor Alan presented to the students a basic, simple cryptographic protocol (PROT_0 as below) and asked students to analyze, give comments on and design

their own versions that they thought were better in some aspect or even could do something more.

PROT_0:

1) A→S: ID_A
2) S→A: Password Request
3) A→S: $H(PW_A)||T$ Where T=Timestamp (of the present time given by the computer clock) PW_A is A's password and H is a cryptographic hash function.

1. Four students volunteered and presented 4 more new versions as below. For your given Z value, consider the according PROT_Z below then *analyze and state the purpose of this protocol and compare it with the original PROT_0 (is there any improvement?)*.

PROT_1: (by student Alice)

1) A→S: ID_A , i.e. A's identity
2) S→A: R, i.e. a random number
3) A→S: $H(PW_A||R)$

PROT_2: (by student Bob)

1) A→S: ID_A
2) S→A: R
3) A→S: $H(PW_A||R)$
4) S→A: $\{k_s\}p_A$ where $p_A= H(PW_A)$

PROT_3: (by student Cathy)

A→S: $ID_A || H(PW_A||T) || T$

Where T= Timestamp

PROT_4: (by student Dave)

1) A→S: ID_A
2) S→A: R
3) A→S: $\{R+1\}p_A$
4) S→A: $\{k_s\}p_A$ where $p_A= H(PW_A)$

2. Instructor Alan then presented the following advanced version and challenged the students to analyze it and find if there is any important remaining issue.

1) A→S: $ID_A || ID_B || r_1$
2) S→A: $\{ID_A || ID_B || r_1 || k_s || \{ID_A || k_s || r_1\} p_B \} p_A$
3) A→B: $\{ID_A || k_s || r_1\} p_B$
4) B→A: $\{r_2\} k_s$
5) A→B: $\{r_1 + r_2\} k_s$

Student Dave successfully responded this challenge by explaining the main issue of this protocol and stating a small but successful fix in the last step (5).

State the purpose of this protocol. Have you seen the issue Dave mentioned? State the required fix in the 5th step if you have also found it and explain.